

ECSE-4040

Digital Electronics

Lecture 21

Bipolar Digital Circuits

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References: Appendix B: HJS

Chapters 5 and 17, Neamen

<https://lms.rpi.edu>

Lecture Outline

- CML implementation
 - Bipolar
 - MOS

High Speed Logic Families

Emitter-Coupled Logic (ECL)

- Bipolar
- III-V FET
- BiCMOS



Current-Mode Logic (CML)

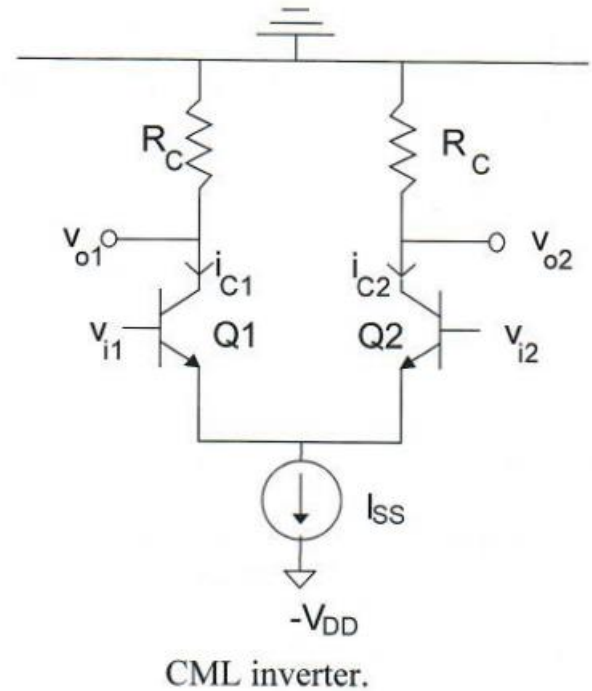
- ☐ Bipolar
- ☐ MOS
- ☐ BiCMOS



**Some high-speed logic families use differential topologies

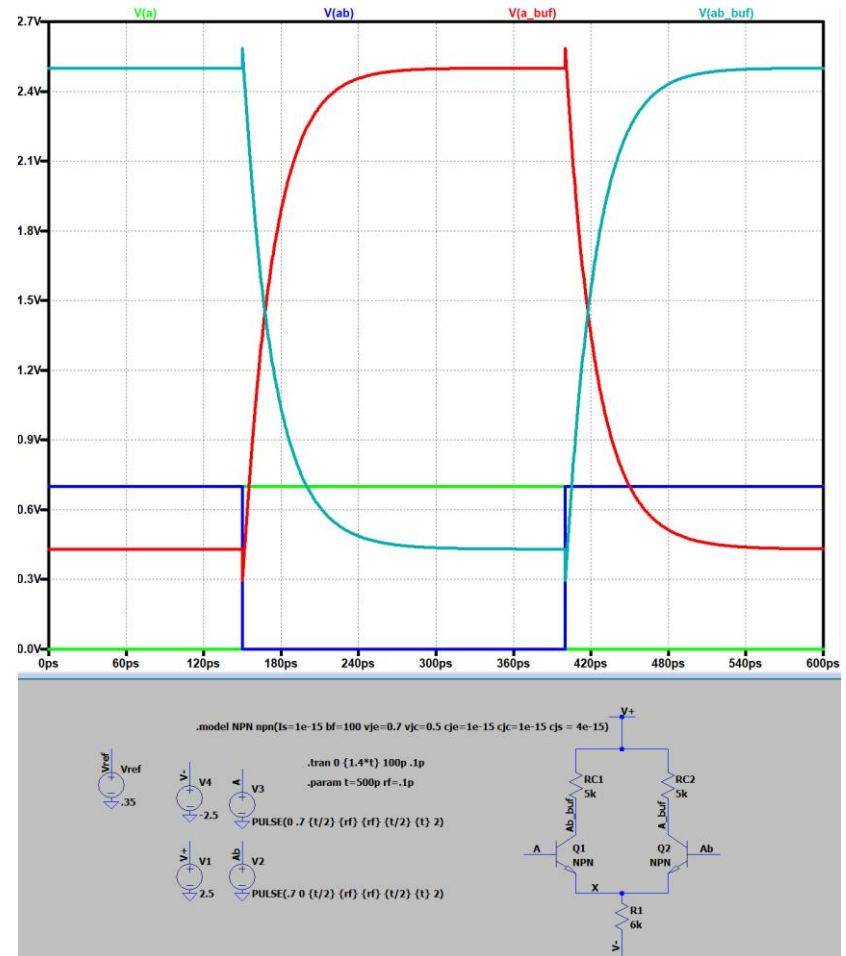
CML

- Current Mode Logic
- ECL can be thought of as an extension to CML.
 - ECL with emitter-followers is a buffered implementation of CML.
 - Low power ECL is a single-rail only implementation.
- CML basics:
 - Fully dual-rail input and output.
 - No reference voltage.
 - Instead read inverted signal.
 - Current steering logic
- Power = $V_{DD} * I_{SS}$
 - Constant

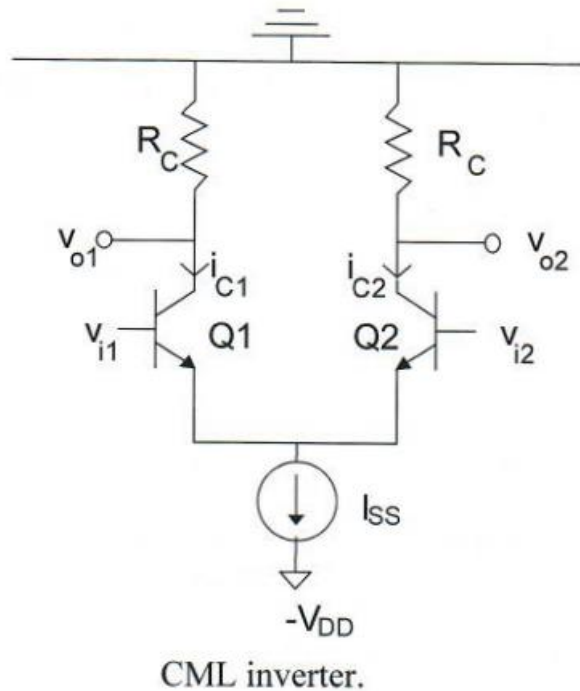


CML Basics

- Do not care for absolute voltage.
- Only voltage difference
- R_C s are now equal



CML Steady-State Voltage



$$V_{i1} < V_{i2} \quad Q_1, OFF \quad Q_2, ON$$

$$V_{o1} = 0$$

$$V_{o2} = -I_{SS} * R_C$$

$$V_E = -I_{SS} * R_C - V_{BE}(FA)$$

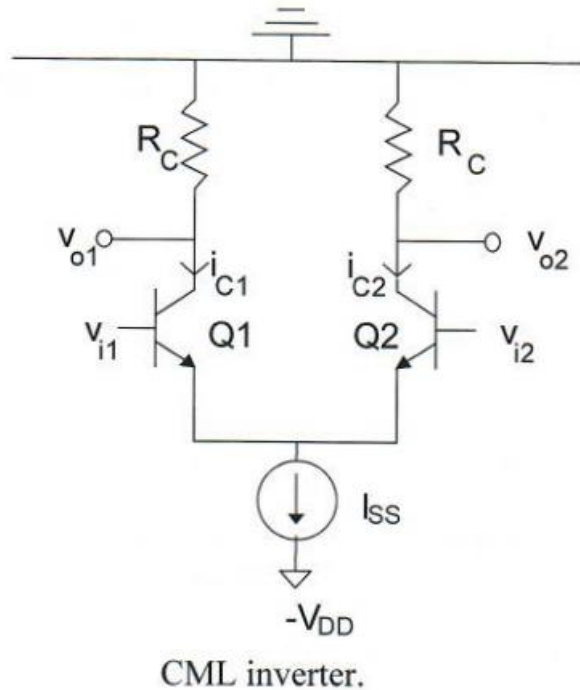
$$V_{i2} < V_{i1} \quad Q_2, OFF \quad Q_1, ON$$

$$V_{o2} = 0$$

$$V_{o1} = -I_{SS} * R_C$$

$$V_E = I_{SS} * R_C - V_{BE}(FA)$$

CML-Basic Concepts



$$\frac{I_{C1}}{I_{C2}} = \frac{e^{\left(\frac{V_{BE1}}{V_T}\right)}}{e^{\left(\frac{V_{BE2}}{V_T}\right)}} = e^{\left(\frac{V_{BE1}-V_{BE2}}{V_T}\right)}$$

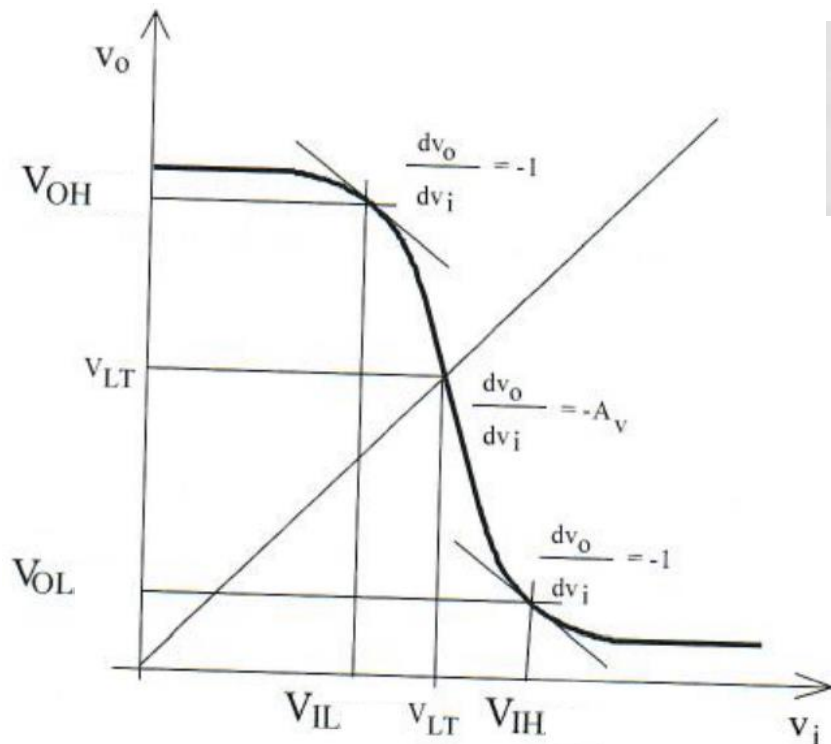
ΔV_{BE}	$\frac{I_{C1}}{I_{C2}}$
0.078	20.08554
0.104	54.59815
0.12	101.0267
0.13	148.4132

CML – Noise Margins

- By definition, $V_{SWING} = V_{OH} - V_{OL}$ (differential)
 - Ideally, $V_{SWING} = 2 * I_{SS} * R_C$
 - V_{IH} , V_{IL} represented as differential voltages
 - V_{IH} is positive, $V_{IL} = -V_{IH}$
- $$V_{IH} \cong V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$
- $$V_{IL} \cong -V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

Ideally,

$$NM = NM_L = NM_H \cong \frac{V_{SWING}}{2} - V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

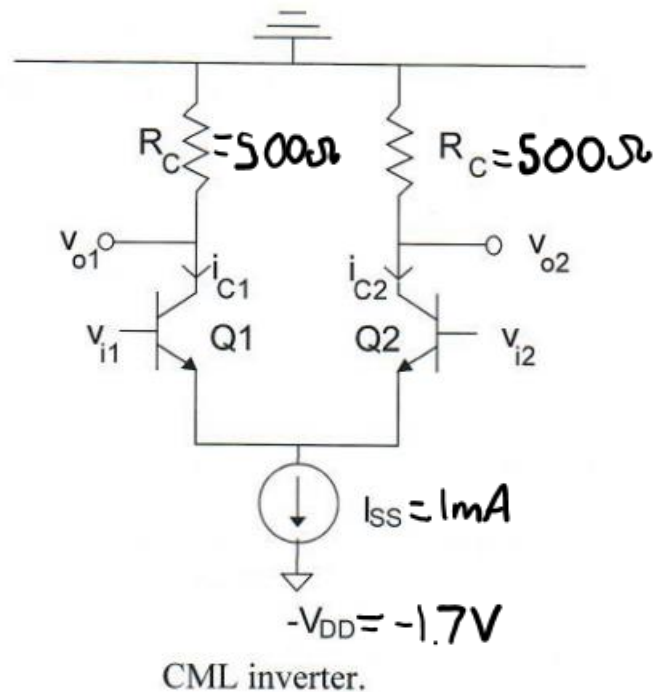


With output loading,

$$NM'_H \cong \left(\frac{V_{SWING}}{2}\right)' - V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

$$\text{where } \left(\frac{V_{SWING}}{2}\right)' = \frac{V_{SWING}}{2} - |V_{OH}'|$$

CML – Noise Margins



$$V_{OL} = -500 \cdot 1\text{m} = -0.5\text{V}$$

$$V_{OH} = 0$$

$$V_{\text{SWING}} = 2 \cdot 1\text{m} \cdot 500 = 1.0\text{V}$$

$$NM = \frac{1.0}{2} - 0.026 \cdot \ln \left[\frac{1.0}{0.026} - 2 \right]$$

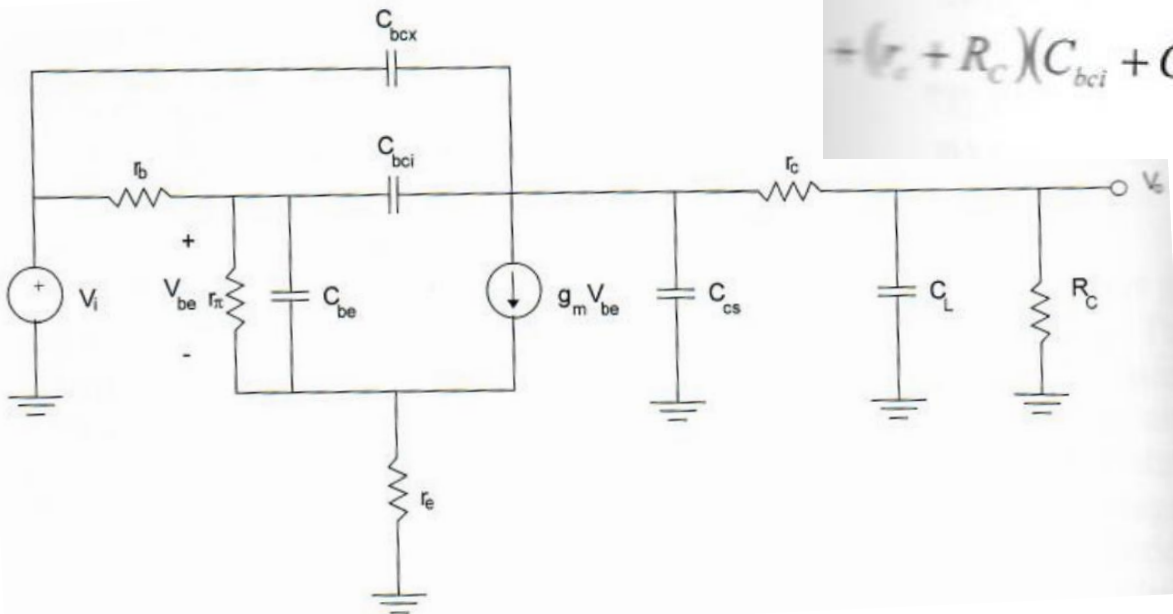
$$NM_L = NM_H = 0.407\text{V}$$

$$R_C = 200\Omega \Rightarrow NM = 0.155\text{V}$$

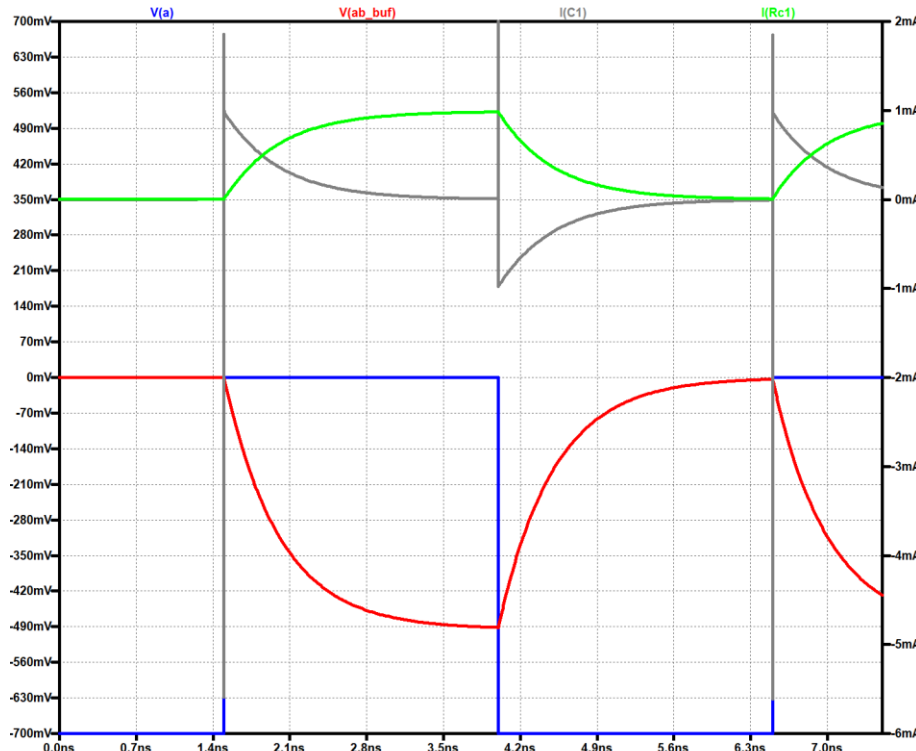
CML Delays

Long form delay:

$$\tau_P = 0.69 \left\{ \frac{r_e + r_b}{1 + g_m r_e} C_{be} + r_b C_{bci} \left[1 + \frac{g_m (r_e + R_C)}{1 + g_m r_e} \right] + (r_e + R_C) (C_{bci} + C_{bcx} + C_{cs}) + R_C C_L \right\}$$



CML Delays



Can we get a simple delay estimate?

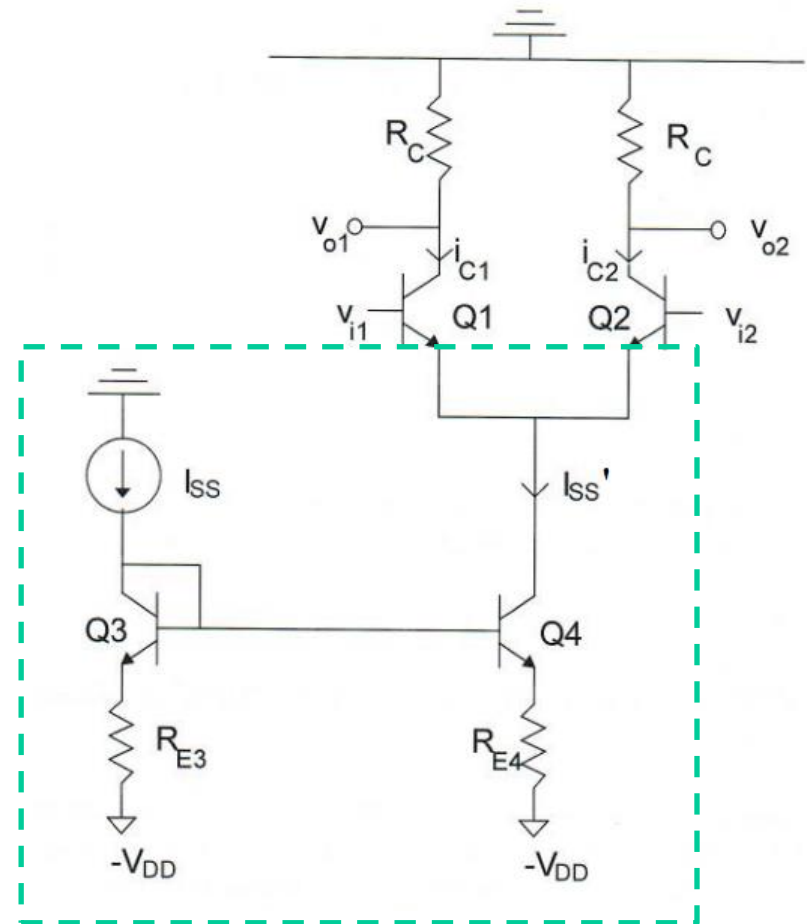
- $\tau_p(\text{simulated}) = 0.35 \text{ ns}$
- $V_{\text{swing}}/2 = 0.25 \text{ V}$
- $C = 1 \text{ pF}$
- $Q = CV = 0.25 \text{ pC}$
- $\tau_p = Q/I$
- $\tau_p = 0.25 \text{ p}/1 \text{ m} = 0.25 \text{ ns}$

Accounting for decline in I_C :

$$\tau_p = 1.41 * C * (V_{\text{swing}}/2) / I_{\text{SS}}$$

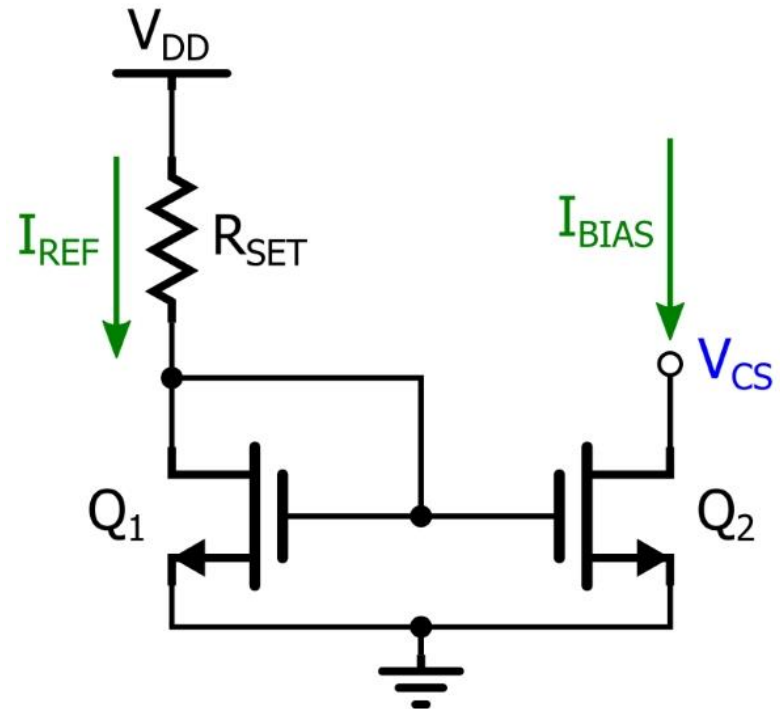
Current Source - BJT

- Varying emitter current is undesirable.
- Replace load dependent R_E with current source
- BJT current source:
 - Forward Active
 - V_C independent
 - $I_{E3} = I_{SS}$
 - Sets V_{E3}
 - $V_{B3} = V_{B4}$
 - R_{E4} sets V_{E4} , V_{BE4}
 - V_{BE4} sets I_{SS}'
 - Problematic – small R_E variance causes major I_{SS}' change



Current Source - MOS

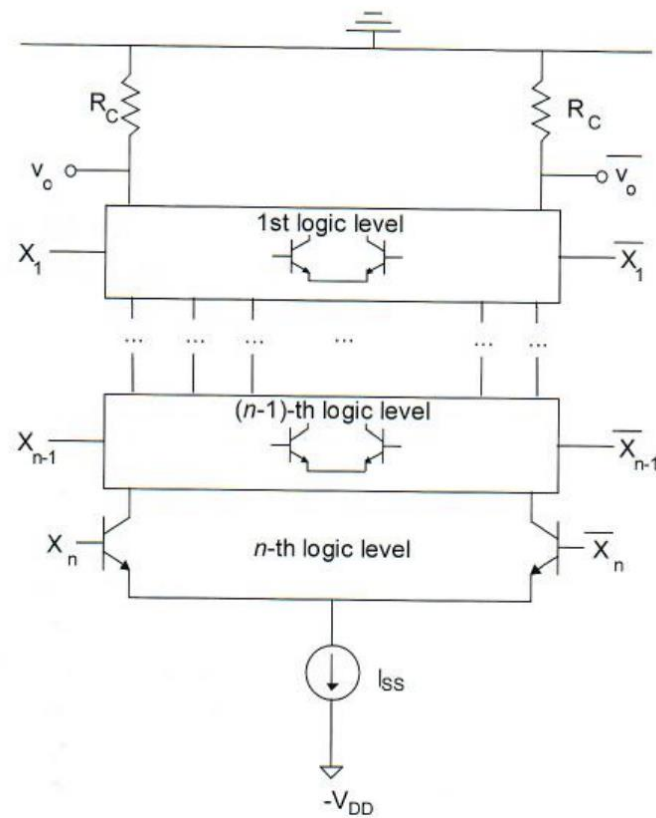
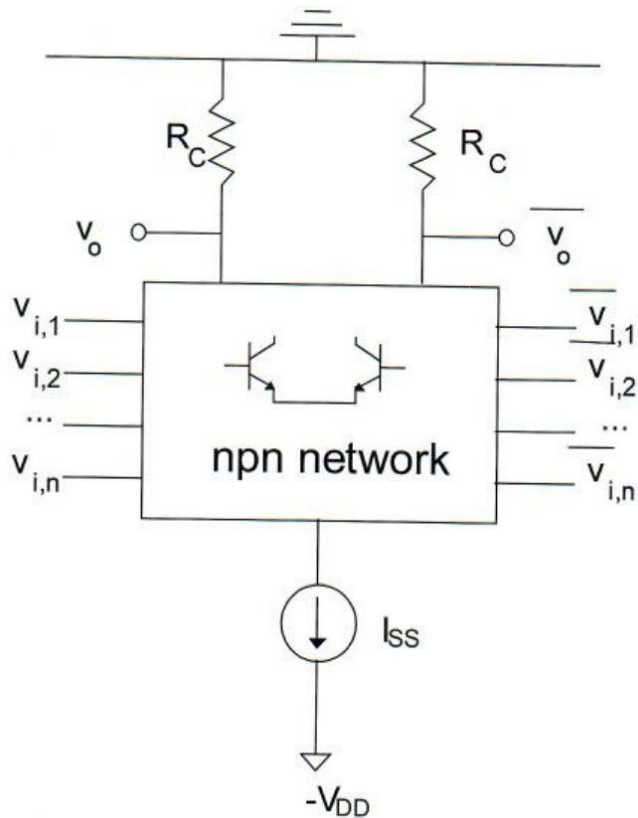
- MOS current source much more stable than BJT current source.
 - Non-exponential gain
 - No secondary resistors required
 - Secondary current varied by W/L ratio.
- Biased for saturation
- Common practice – Use MOS current mirrors to set biasing currents.
 - Even for BJT circuits



<https://www.allaboutcircuits.com/technical-articles/the-basic-mosfet-constant-current-source/>

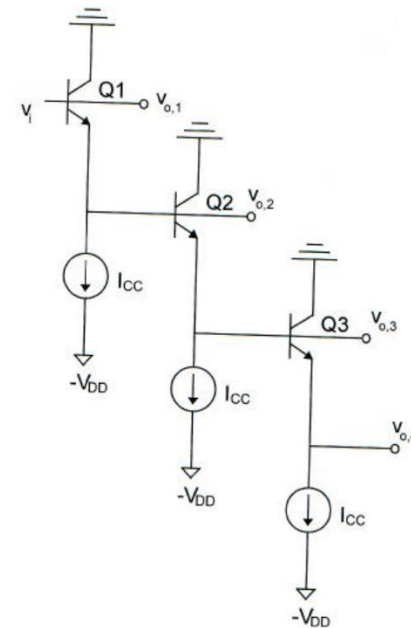
Cascading Logic

- Multiple input voltage tiers
 - Similar to ECL
- Current steering:
 - Have current = logic 1
 - No current = logic 0

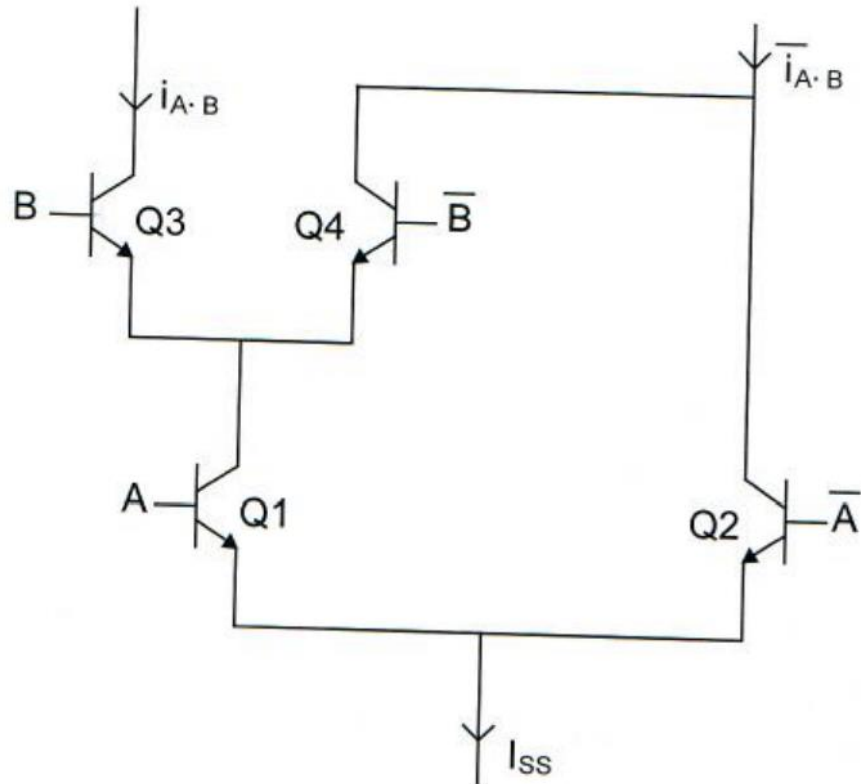


Cascading Logic

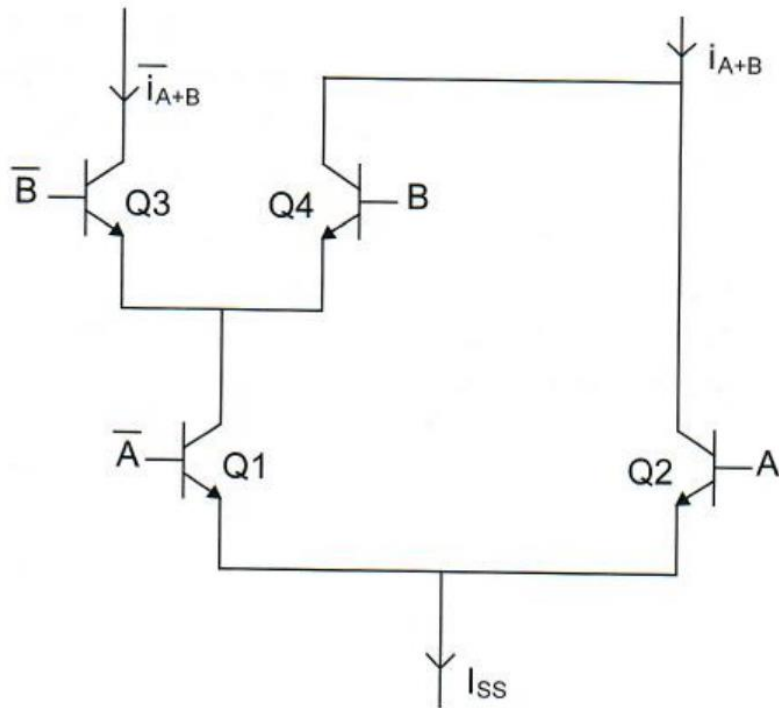
- Need to downstep input voltages for lower tiers
 - Similar to ECL



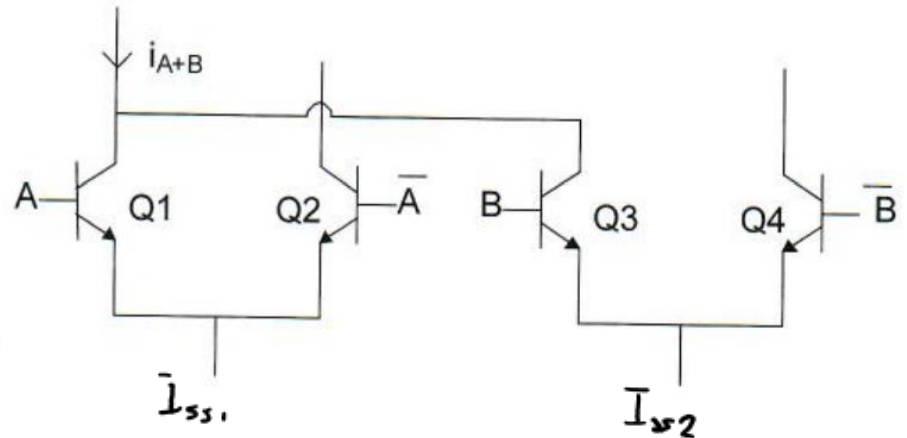
Basic Logic Functions - AND



Basic Logic Functions - OR



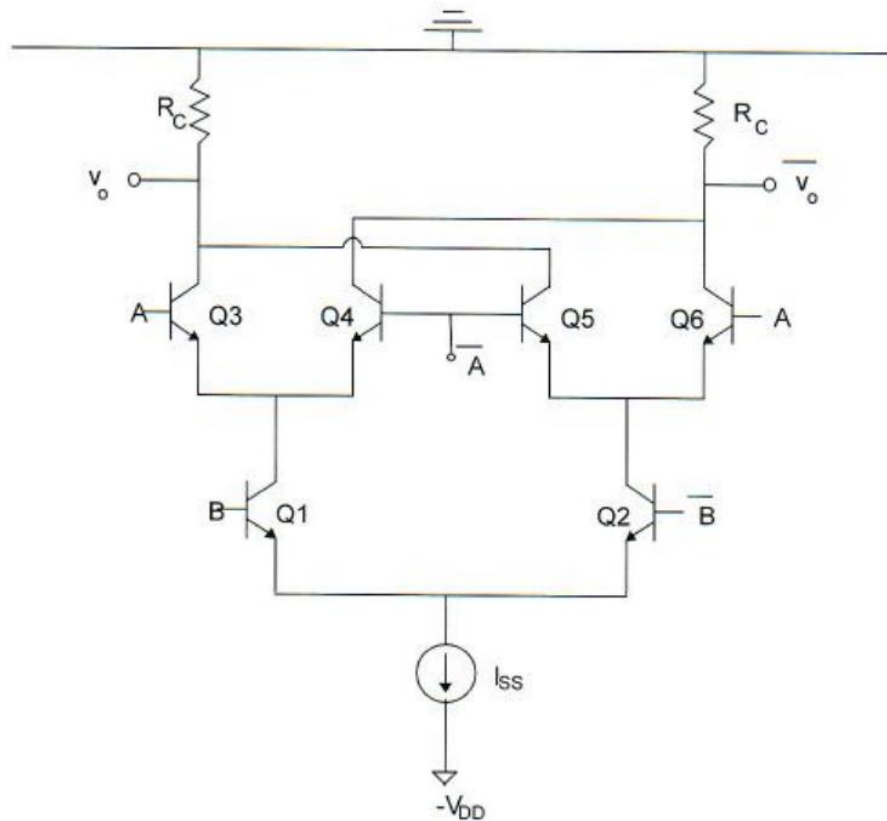
- Stacked



- Non-Stacked

Basic Logic Functions

- Try this one!



Complex Logic Functions

- Remember, the NPN BJTs are active pull-down.
- Build F function with positive inputs
 - Having current means low voltage
 - Can swap output names for correct output.
 - Double negative
- Start at the bottom, work up
 - Ex: start at base with X_4
 - Branch up to X_3 input, repeat

TABLE 3.1

$X_1 X_2 X_3 X_4$	associated minterm	$F(X_1, X_2, X_3, X_4)$	$\overline{F(X_1, X_2, X_3, X_4)}$
0000	$\overline{X_1} \overline{X_2} \overline{X_3} \overline{X_4}$	0	1
0001	$\overline{X_1} \overline{X_2} \overline{X_3} X_4$	0	1
0010	$\overline{X_1} \overline{X_2} X_3 \overline{X_4}$	1	0
0011	$\overline{X_1} \overline{X_2} X_3 X_4$	0	1
0100	$\overline{X_1} X_2 \overline{X_3} \overline{X_4}$	0	1
0101	$\overline{X_1} X_2 \overline{X_3} X_4$	0	1
0110	$\overline{X_1} X_2 X_3 \overline{X_4}$	0	1
0111	$\overline{X_1} X_2 X_3 X_4$	0	1
1000	$X_1 \overline{X_2} \overline{X_3} \overline{X_4}$	0	1
1001	$X_1 \overline{X_2} \overline{X_3} X_4$	1	0
1010	$X_1 \overline{X_2} X_3 \overline{X_4}$	0	1
1011	$X_1 \overline{X_2} X_3 X_4$	0	1
1100	$X_1 X_2 \overline{X_3} \overline{X_4}$	1	0
1101	$X_1 X_2 \overline{X_3} X_4$	1	0
1110	$X_1 X_2 X_3 \overline{X_4}$	1	0
1111	$X_1 X_2 X_3 X_4$	1	0

Complex Logic Functions

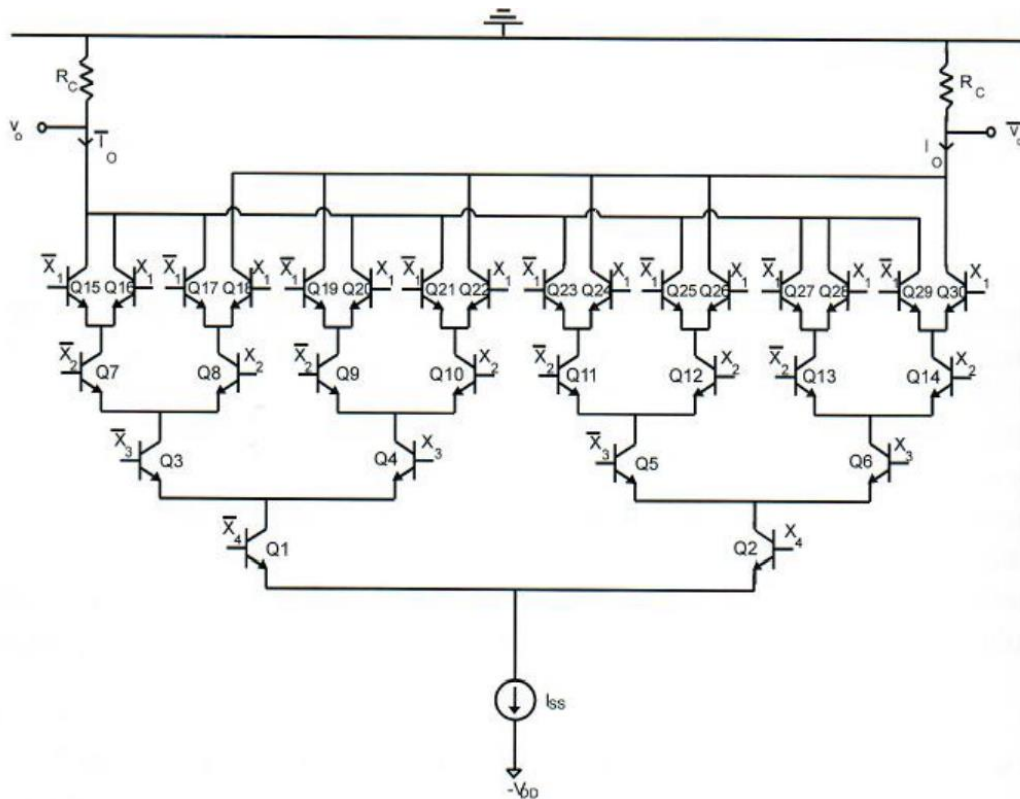


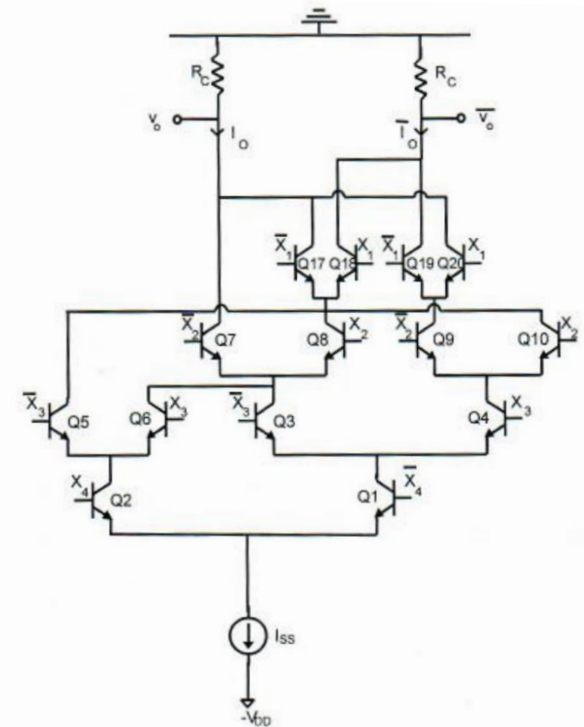
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0011	$\overline{X_1} \overline{X_2} X_3 X_4$	0	1
0100	$\overline{X_1} X_2 \overline{X_3} \overline{X_4}$	0	1
0101	$\overline{X_1} X_2 \overline{X_3} X_4$	0	1
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0111	$\overline{X_1} X_2 X_3 X_4$	0	1
1000	$X_1 \overline{X_2} \overline{X_3} \overline{X_4}$	0	1
1001	$X_1 \overline{X_2} \overline{X_3} X_4$	1	0
1010	$X_1 \overline{X_2} X_3 \overline{X_4}$	0	1
1011	$X_1 \overline{X_2} X_3 X_4$	0	1
1100	$X_1 X_2 \overline{X_3} \overline{X_4}$	1	0
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Complex Logic Functions

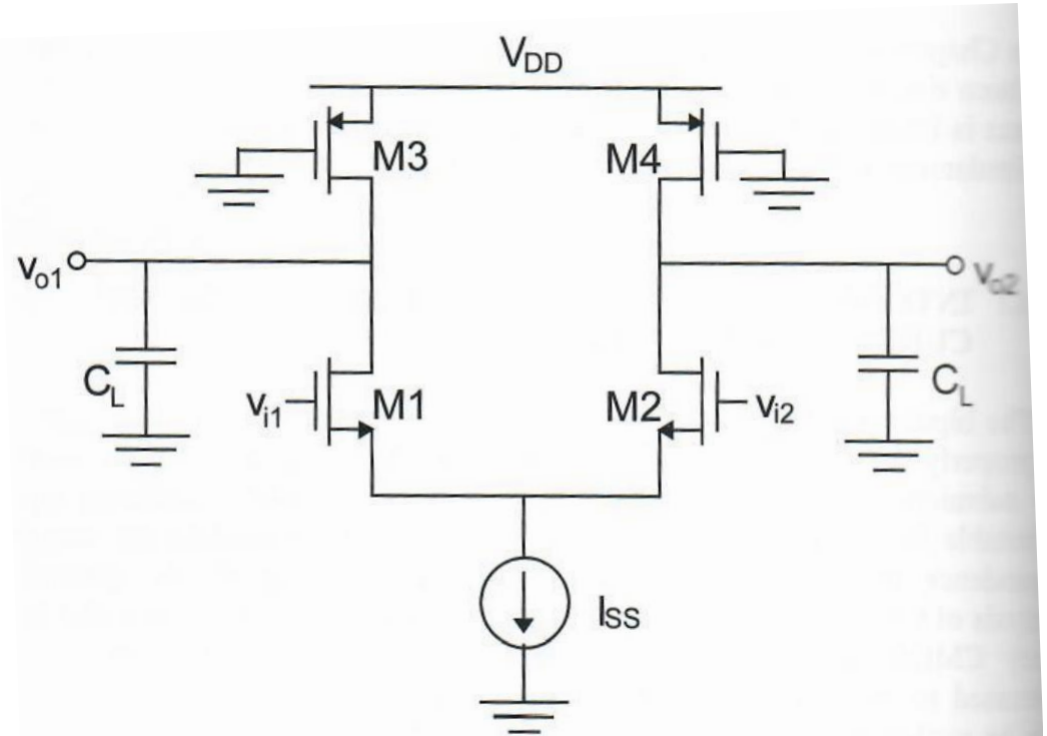
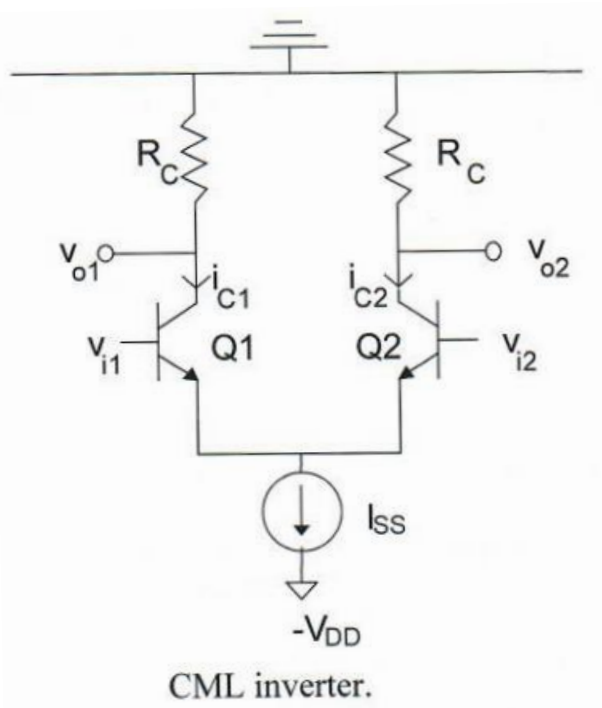
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1000	$X_1 \overline{X_2} \overline{X_3} \overline{X_4}$	0	1
1001	$X_1 \overline{X_2} \overline{X_3} X_4$	1	0
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1011	$X_1 \overline{X_2} X_3 X_4$	0	1
1100	$X_1 X_2 \overline{X_3} \overline{X_4}$	1	0
1101	$X_1 X_2 \overline{X_3} X_4$	1	0
1110	$X_1 X_2 X_3 \overline{X_4}$	1	0
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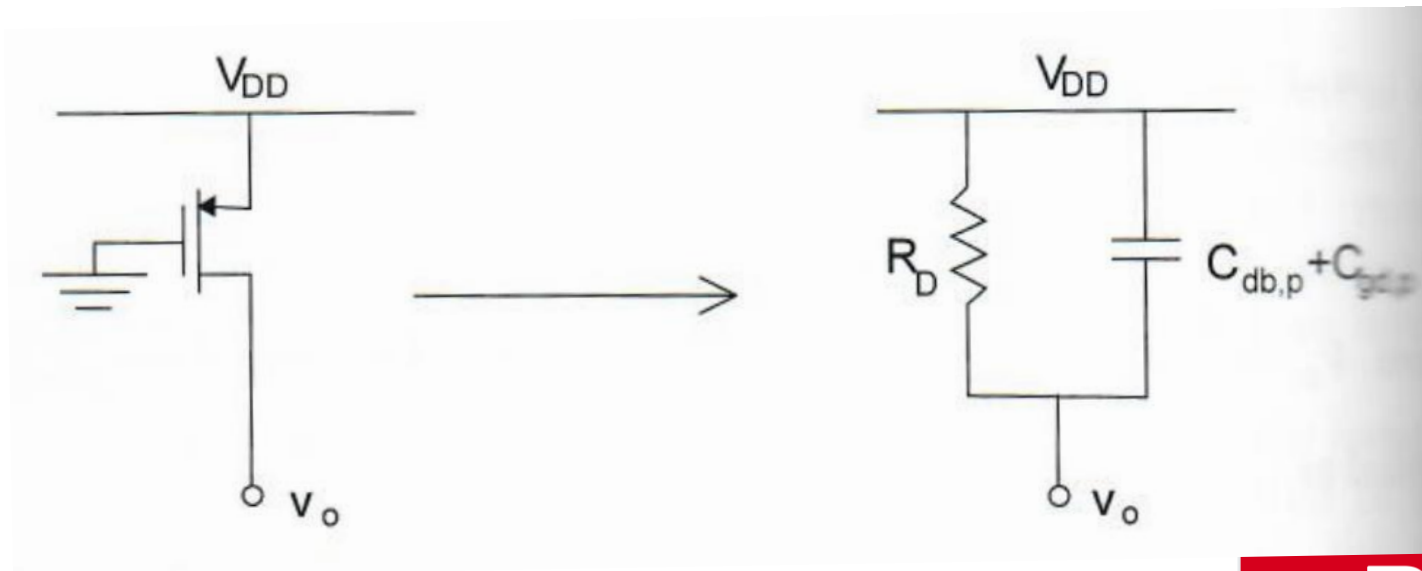
CML - MOS

- BJT difficult to integrate on same silicon die.
- CML also works with MOSFETs
 - CML is best with BJT for high gain.
 - CML with MOS sometimes called SCL (Source Coupled Logic)



Load Resistor

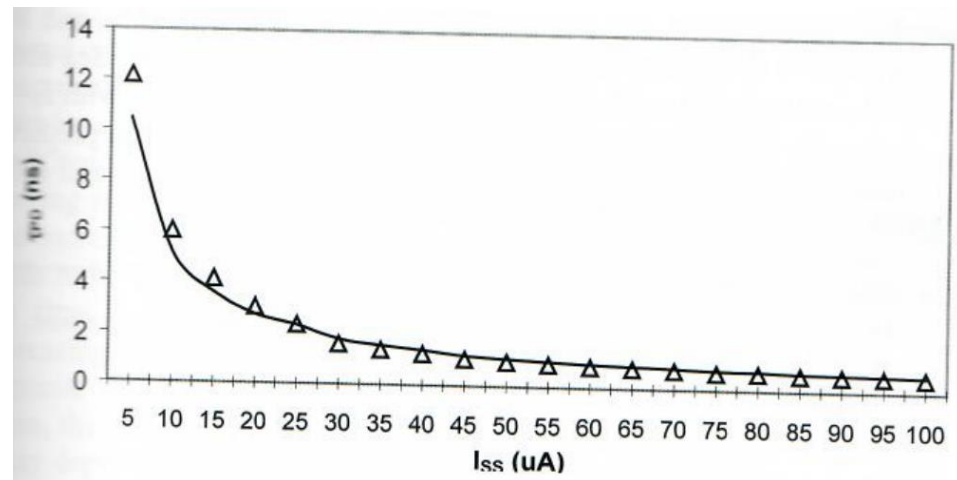
- MOS – Can use linearly biased PMOS as load resistor.
 - BJT CML can as well, and often does.
 - $R_D = V_{DS}/I_{DS}(\text{linear})$
 - Can solve V_{DS} with $I_{DS} = I_{SS}$
 - $V_{OH} = V_{DD}$
 - $V_{OL} = V_{DD} - R_D * I_{SS}$



Delays

- Use small signal model for accurate delays
 - Valid since MOSFETs biased to always be saturated
 - Small voltage swing

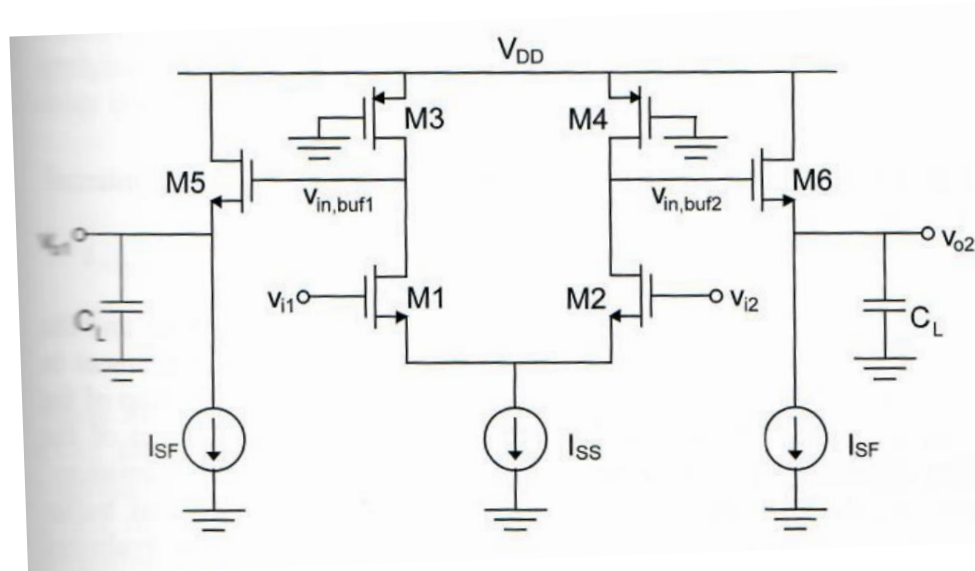
$$\tau_{PD,buf} = 0.69 \left[R_D \left(C_{gd,buf} + C_{gs,buf} \frac{\frac{g_{mb,buf}}{g_{m,buf}}}{1 + \frac{g_{mb,buf}}{g_{m,buf}}} \right) + \frac{C_L + C_{gs,buf}}{g_{m,buf}} \right]$$



- Alternatively, can approximate
 - $\tau_p = 1.41 * C * (V_{swing}/2) / I_{SS}$

$$C_L = 200 \text{ fF}$$

Buffering



- Can reduce delay with output buffers.

CML MOS – Logic Gates

- XOR
- Same method as with BJTs
- Power still $I_{SS} * V_{DD}$

